

Barnes maze report

Exported 2026-09-08 · 0.1.0

3 scored trials from 3 animals. 0 not yet scored.

Methods

Behaviour was recorded at 29.929755043227665 frames per second on a 92 cm diameter Barnes maze platform with 20 holes. Videos were scored using Barnes maze scorer v0.1.0.

Animal position was tracked and hole investigations were defined as the nose remaining within 3 cm of a hole centre for at least 0.2 s, with visits to the same hole within 1 s merged into a single investigation.

Primary latency was the time to first come within 5 cm of the target hole. Escape was scored as a disappearance at the target hole persisting for at least 8 frames, corroborated by the last observed position. Errors were counted as distinct non-target holes investigated.

Frames in which the animal could not be localised and for which there was no evidence of hole entry were excluded rather than interpolated; the proportion of successfully localised frames is reported per trial. Trajectories were median filtered over 5 frames.

Search strategy was classified as spatial, serial or random using path directness, the sequence of hole visits and target quadrant occupancy, following the feature set of Illouz et al. (2016). Classifications were reviewed and could be overridden by an experimenter; the source of each label is recorded in the dataset.

A total of 3 trials from 3 animals were analysed.

Check this paragraph against your protocol before use. It reports the settings this analysis used, not a substitute for describing your own experiment.

Parameters

parameter	value
tool_version	0.1.0
gap_handling	skipped, not interpolated
investigationRadiusCm	3
investigationMinDwells	0.2
investigationRefractoryS	1
escapeConfirmFrames	8
reachedTargetRadiusCm	5
smoothingWindowFrames	5
maxGapFillFrames	0
trialTimeoutS	180

Measurements

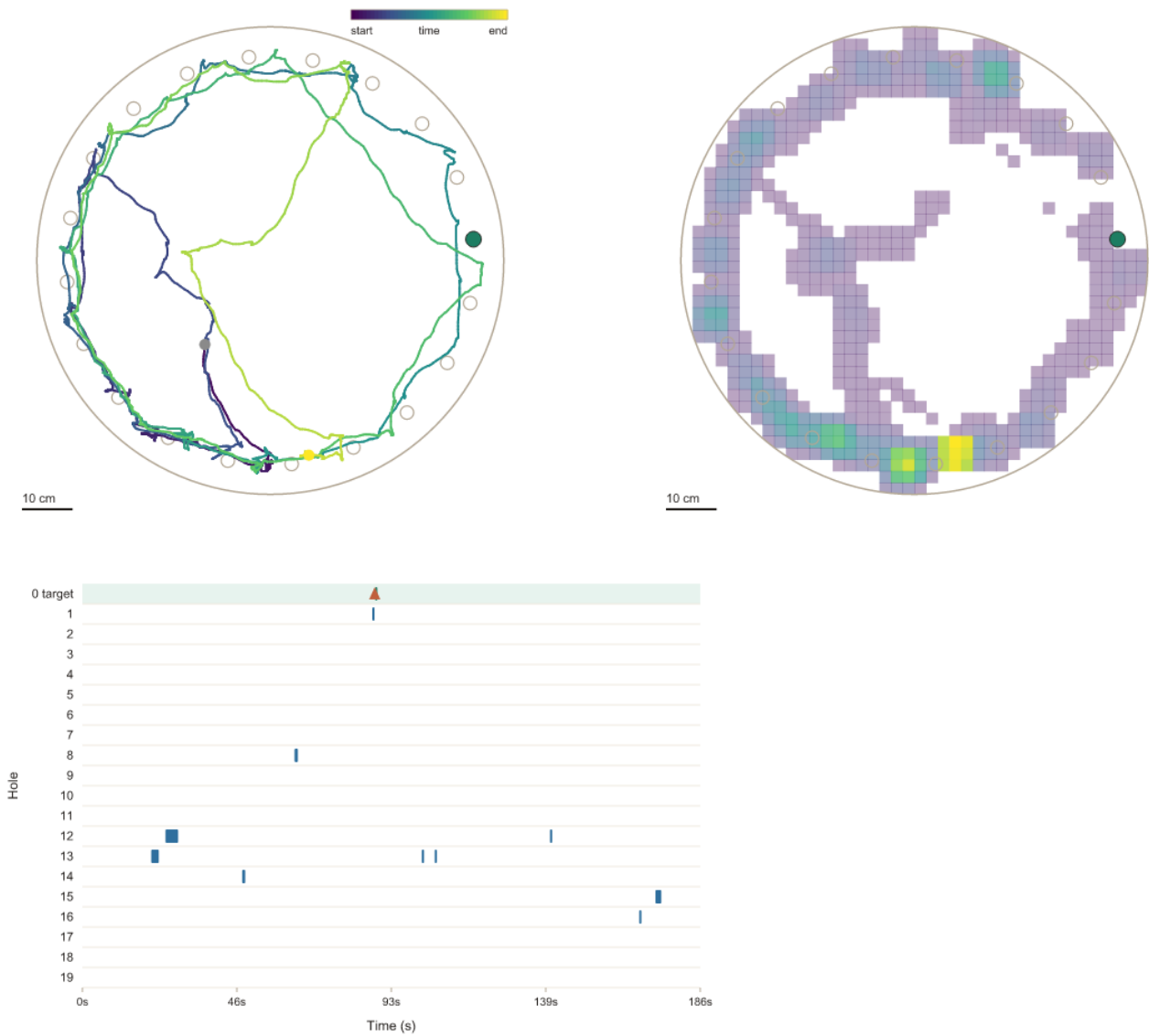
animal	cohort	day	reached	lat_s	err	path_cm	strategy
test50	sample		yes	87.8	5	737	random
test51	sample		no		2	145	random
test53	sample		no		4	41	random

Events

animal	event	hole	start_s	end_s	source
test50	investigation	13	20.65	22.95	auto
test50	investigation	12	24.96	28.77	auto
test50	investigation	14	47.98	48.91	auto
test50	investigation	8	63.72	64.68	auto
test50	investigation	1	87.07	87.44	auto
test50	reached-target	0	87.77	87.77	auto
test50	investigation	0	87.91	88.11	auto
test50	investigation	13	101.94	102.44	auto
test50	investigation	13	105.75	106.38	auto
test50	investigation	12	140.40	140.83	auto
test50	investigation	16	167.19	167.39	auto
test50	investigation	15	172.07	173.84	auto
test51	investigation	8	13.33	22.93	auto
test51	investigation	1	37.39	48.32	auto
test53	investigation	19	5.01	8.72	auto
test53	investigation	18	11.89	14.06	auto
test53	investigation	16	17.87	20.95	auto
test53	investigation	17	24.15	24.42	auto
test53	investigation	18	27.19	30.27	auto

test50 · test50.mp4

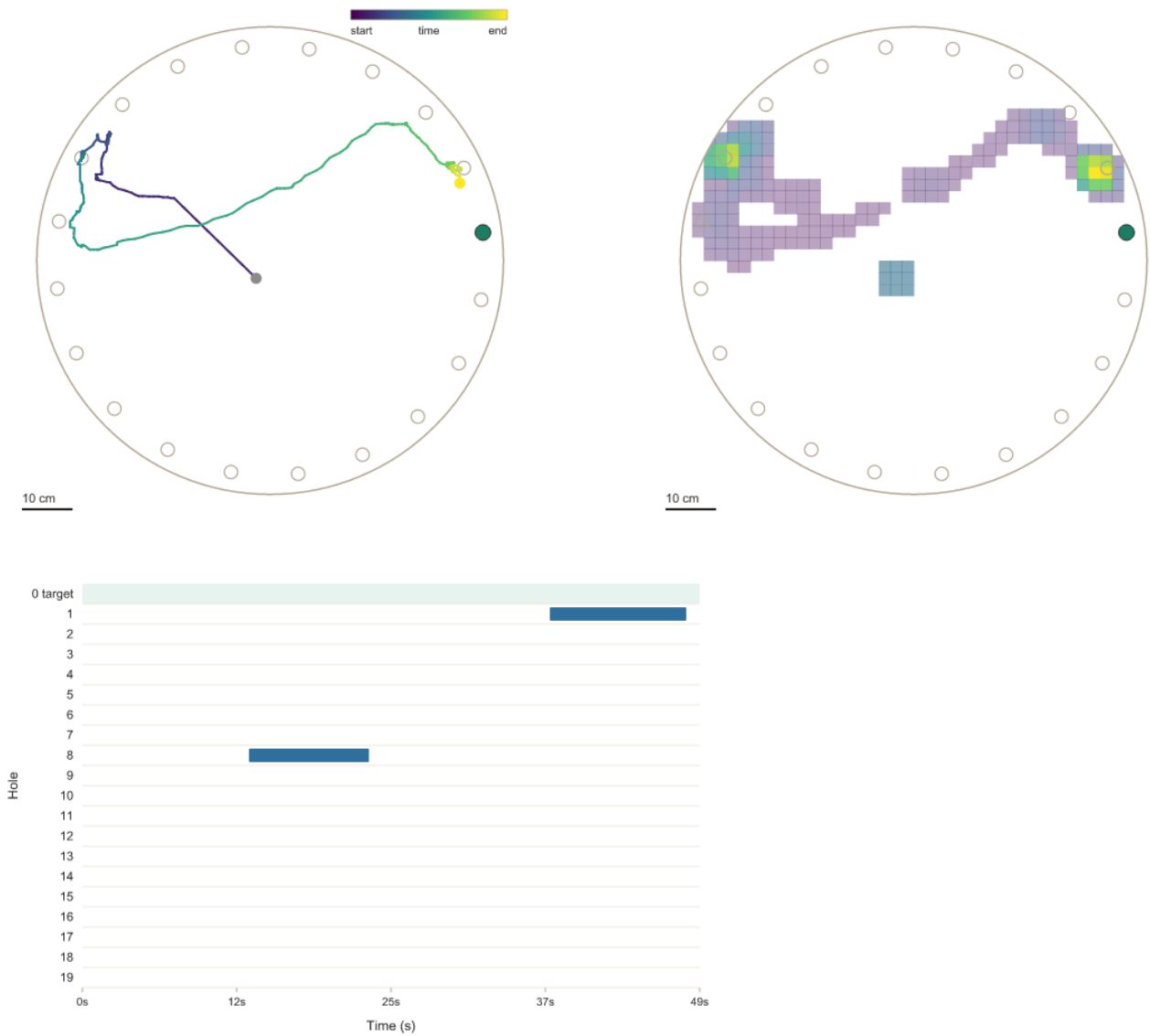
Day — sample · latency 87.77 s · errors 5 · random · tracked 97% · A 5.0s stretch with no detection. Check whether the animal was in a hole.



Time-colored path, occupancy heatmap, hole-visit raster.

test51 · test51.mp4

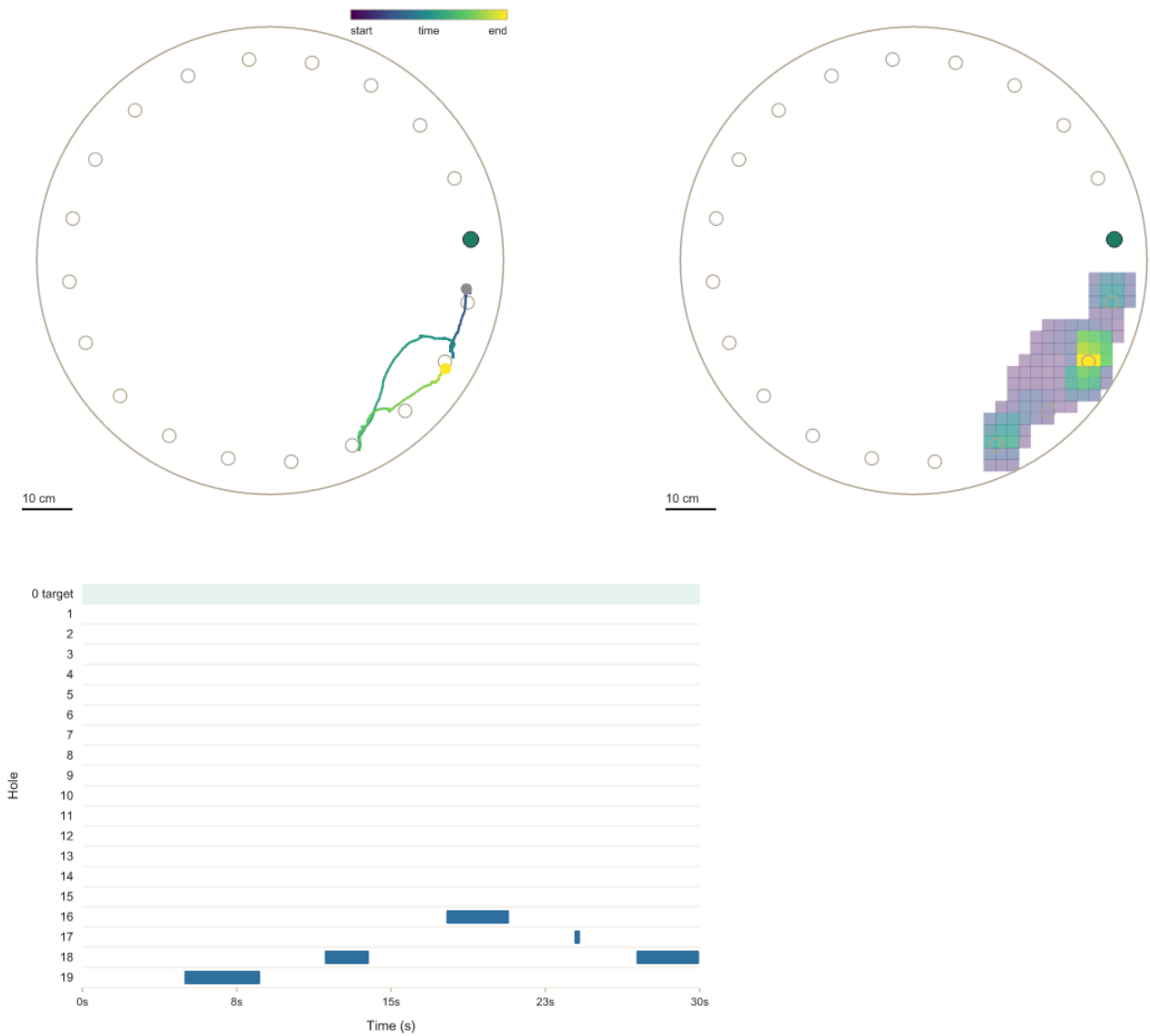
Day — sample · latency never reached · errors 2 · random · tracked 100%



Time-colored path, occupancy heatmap, hole-visit raster.

test53 · test53.mp4

Day — sample · latency never reached · errors 4 · random · tracked 83% · A 5.0s stretch with no detection. Check whether the animal was in a hole.



Time-colored path, occupancy heatmap, hole-visit raster.